

VLSI Design Internship - Task 3

Verilog RTL Design of Sequential Circuits and Flip-Flops

Objective

To understand sequential circuits, flip-flops, registers, counters, Verilog RTL coding, testbench creation, and waveform analysis.

Introduction

Sequential circuits depend on current inputs and previous states and operate using clock signals.

Implemented Designs

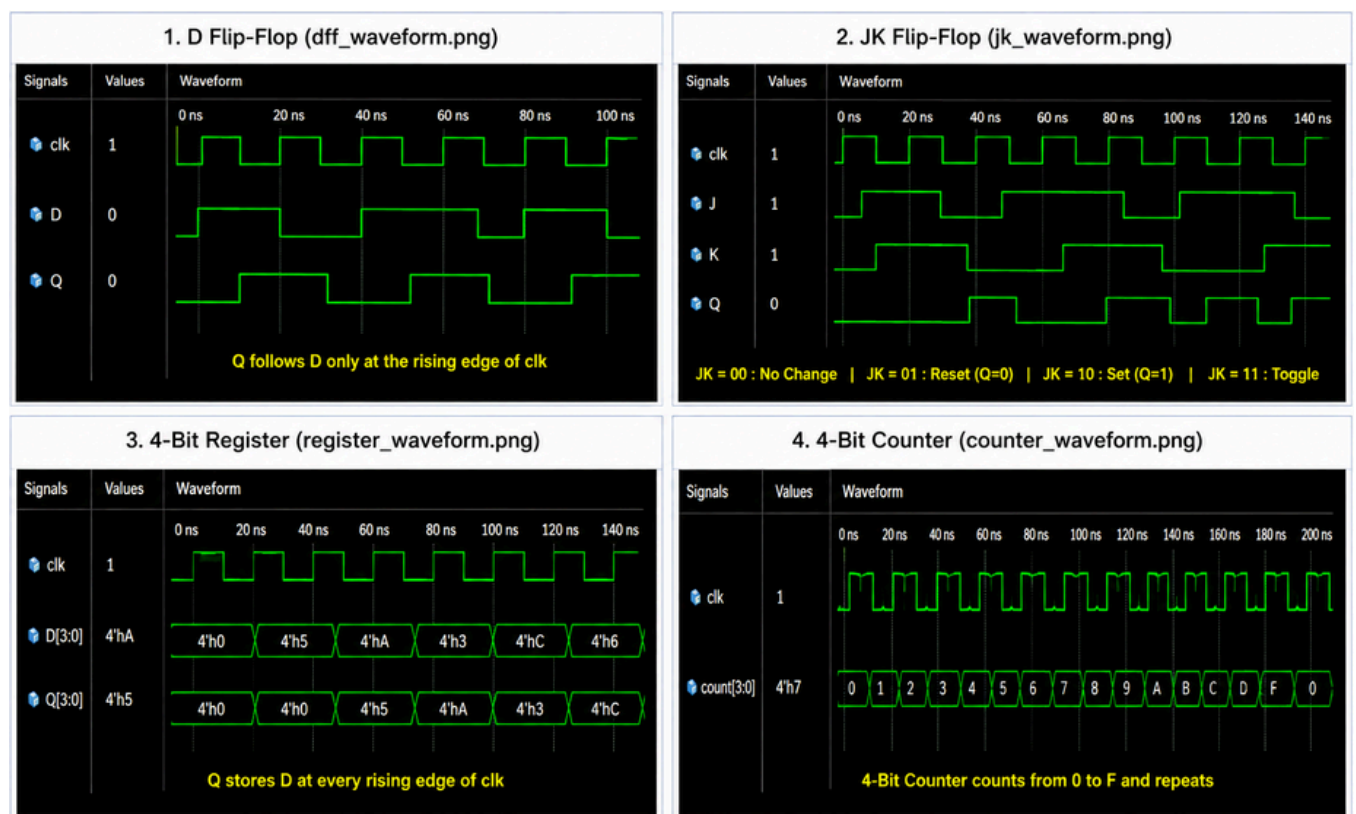
DFlip-Flop, JKFlip-Flop, 4-Bit Register, and 4-Bit Binary Counter.

Tools Used

EDA Playground, Icarus Verilog, and GTKWave.

Simulation Results

All circuits were simulated using Verilog testbenches and verified through waveform analysis.



Note: All simulations are done using EDA Playground (Icarus Verilog + GTKWave)

Time period = 20ns, Clock high = 10ns, Clock low = 10ns

Conclusion

Task3 provided practical experience in RTL-based sequential circuit design using Verilog HDL.